

ITSimplera Institute Internship

Week 1 Task Report

Exploratory Data Analysis Report

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Objective

This report explores customer purchase behavior in the Online Retail II dataset (UK-based online retailer, 2009–2011) to help the company evaluate readiness for a recommendation system. It covers dataset structure, data quality issues, product and country performance, revenue trends, correlations, and outliers.

Dataset Overview

- **Source:** UCI Online Retail II (Excel file with two sheets: "Year 2009-2010" and "Year 2010-2011")
- **Combined shape:** 1,067,371 rows × 8 columns
- **Columns:** Invoice, StockCode, Description, Quantity, InvoiceDate, Price, Customer ID, Country

```
Shape: (1067371, 8)
Columns: ['Invoice', 'StockCode', 'Description', 'Quantity', 'InvoiceDate', 'Price', 'Customer ID', 'Country']
Invoice      object
StockCode    object
Description   object
Quantity      int64
InvoiceDate  datetime64[ns]
Price         float64
Customer ID   float64
Country       object
dtype: object
```

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	2009-12-01 07:45:00	6.95	13085.0	United Kingdom
1	489434	79323P	PINK CHERRY LIGHTS	12	2009-12-01 07:45:00	6.75	13085.0	United Kingdom
2	489434	79323W	WHITE CHERRY LIGHTS	12	2009-12-01 07:45:00	6.75	13085.0	United Kingdom
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	2009-12-01 07:45:00	2.10	13085.0	United Kingdom
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	2009-12-01 07:45:00	1.25	13085.0	United Kingdom

Data Quality Findings

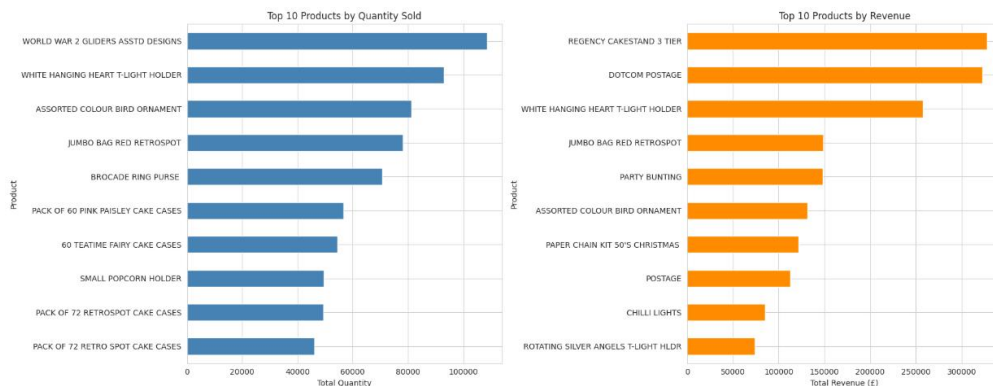
Issue	Value
Missing Description	4,382 rows (0.41%)
Missing Customer ID	243,007 rows (22.77%)
Duplicate rows	34,335 (3.22%)
Negative Quantity (likely returns)	22,950 rows
Zero/negative Price	6,207 rows
Cancelled invoices (prefix "C")	19,494

```
missing_count  missing_pct
Description      4382         0.41
Customer ID    243007        22.77
Duplicate rows: 34335 (3.22%)
Negative Quantity rows: 22950
Zero/negative Price rows: 6207
Cancelled invoices (start with C): 19494
```

Takeaway: Nearly a quarter of transactions can't be tied to a customer, and returns/cancellations are a meaningful part of the data, not noise.

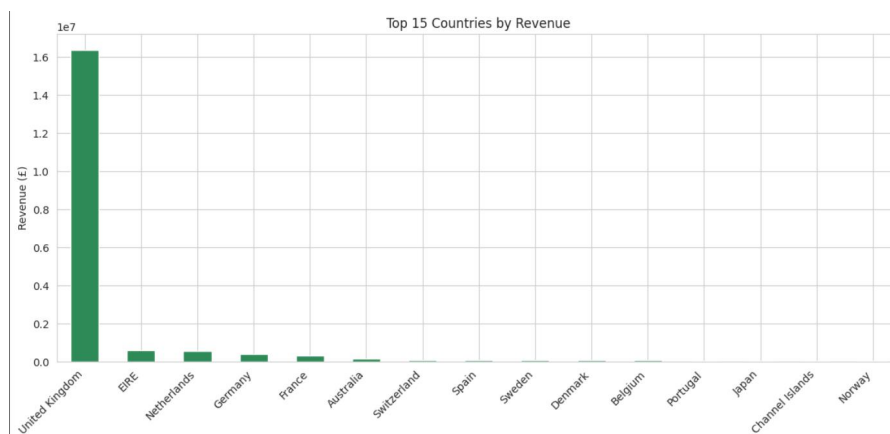
Top-Selling Products

- **By quantity:** WORLD WAR 2 GLIDERS ASSTD DESIGNS (108,545), WHITE HANGING HEART T-LIGHT HOLDER (93,050), ASSORTED COLOUR BIRD ORNAMENT (81,306), and others.
- **By revenue:** REGENCY CAKESTAND 3 TIER (£327,814), DOTCOM POSTAGE (£322,647), WHITE HANGING HEART T-LIGHT HOLDER (£257,534), and others.



Sales by Country

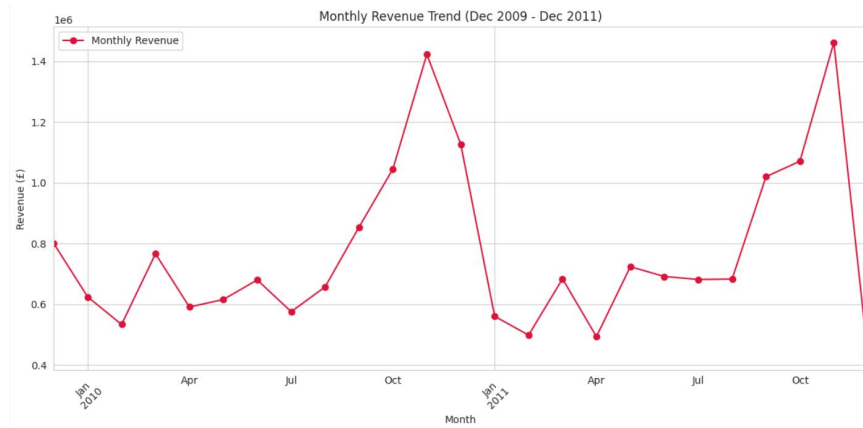
United Kingdom generates **£16.38M**, or **84.94%** of total revenue. All other countries trail far behind, EIRE (£615K), Netherlands (£548K), Germany (£418K), France (£328K).



Takeaway: The business is heavily UK-dependent; international markets are comparatively under-penetrated.

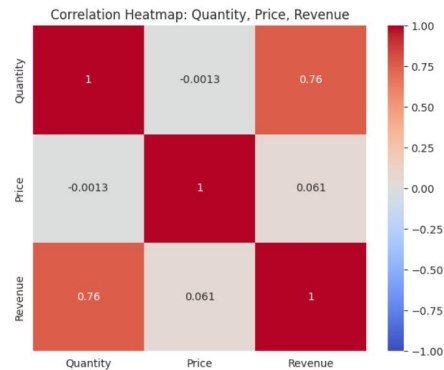
Monthly Revenue Trend

Revenue shows a clear seasonal pattern with peaks around Q4 (Oct–Dec) each year, consistent with holiday shopping, and dips in other months.



Correlation Analysis

	Quantity	Price	Revenue
Quantity	1.00	-0.001	0.76
Price	-0.001	1.00	0.06
Revenue	0.76	0.06	1.00

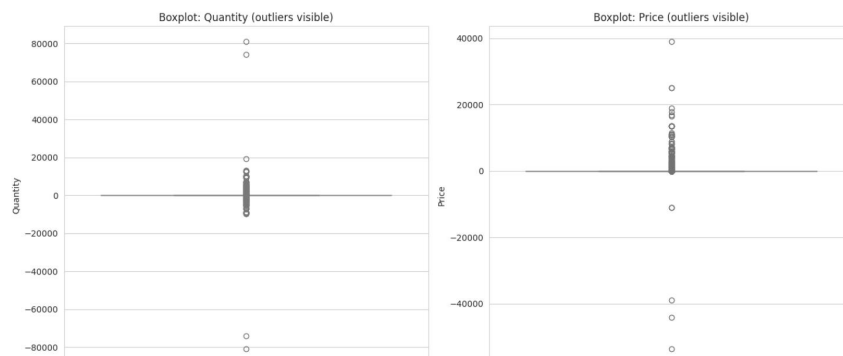


Takeaway: Revenue is driven mainly by quantity sold, not unit price — customers buy many low-cost items rather than fewer expensive ones.

Outlier Detection

Using the IQR method:

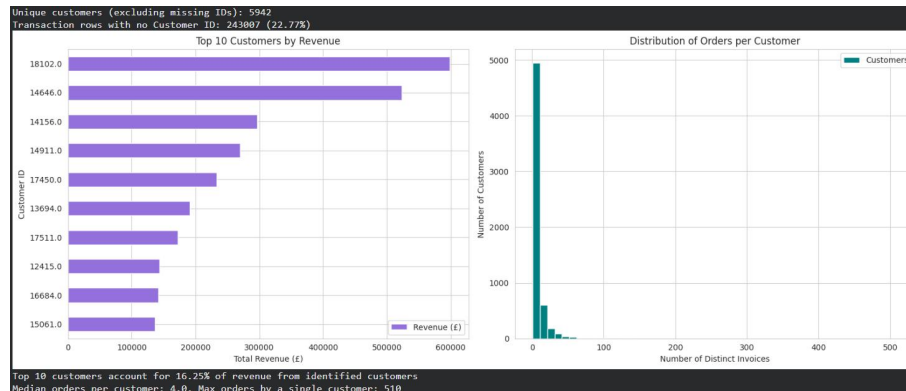
- **Quantity:** 116,489 outliers (normal range: -12.5 to 23.5)
- **Price:** 68,105 outliers (normal range: -3.10 to 8.50)



Takeaway: A large number of extreme values suggests a mix of bulk/wholesale orders and possible data entry errors, both are worth investigating before modeling.

Customer Analysis

- Unique identified customers: 5,942
- Transactions with no Customer ID: 243,007 (22.77%)
- Top 10 customers' share of identified revenue: 16.25%
- Median orders per customer: 4 | Max orders by one customer: 510



Takeaway: A small group of high-frequency buyers (likely wholesale/business accounts) contributes disproportionately to revenue, alongside a much larger base of low-frequency shoppers.

Business Insights

1. **Data quality needs attention before modeling:** 22.77% of rows have no Customer ID and 3.22% are exact duplicates; a recommendation system would lose nearly a quarter of transactions unless this is resolved.
2. **Cancellations and returns are material, not noise:** ~19,500 cancelled invoices and ~22,900 negative-quantity rows likely reflect real customer dissatisfaction and should be modeled as a distinct signal.
3. **Revenue is heavily product-concentrated:** a small set of SKUs dominate sales, so a recommender should balance surfacing bestsellers with promoting long-tail discovery.
4. **The business is highly dependent on the UK market:** 84.94% of revenue comes from one country, representing both a concentration risk and a growth opportunity abroad.
5. **Revenue is seasonal:** clear Q4 peaks mean demand forecasting and recommendations should account for time-of-year effects.
6. **A small group of high-value customers drives disproportionate revenue:** top 10 customers account for 16.25% of identified revenue, suggesting wholesale/business accounts need separate treatment from casual shoppers.
7. **Quantity and Price contain heavy outliers:** over 180,000 combined outlier rows point to bulk orders and possible data entry errors that should be investigated (not removed) before further modeling.

Conclusion

The dataset is rich enough to support a recommendation system but requires a clear data-cleaning strategy, particularly around missing Customer IDs, cancellations, and postage/non-product line items, before it can be used reliably for personalization.